



QUESTION BANK

FUNDAMENTALS OF QUANTUM COMPUTING

UNIT I

1. Define Rayleigh-jeans law?

According to Rayleigh-jean's the radiation emitted by the black body at a temperature is directly proportional to absolute temperature of black body and inversely proportional to fourth power of the wavelength.

$$E \propto T$$

$$E(u) \propto \frac{1}{\lambda^4}$$

$$E(u) = \frac{8\pi kT}{\lambda^4}$$

2. Define Stephen-Boltzmann law?

Stephen-Boltzmann law is the relation between the radiation emitted by the black body to the temperature.

The total radiation emitted by the black body is directly proportional to the fourth power of absolute temperature of black body.

$$\text{i.e., } E \propto T^4$$

$$E = \sigma T^4$$

$$\sigma = 5.67 \times 10^{-8} \text{ w/m}^2 - \text{K}^4.$$

3. Write the expression for the Weins law?

It explains a shift of peak wavelength with the temperature.

The peak wavelength emitted by the black body temperature is inversely proportional to the absolute temperature of black body.

$$\text{i.e., } \lambda_m \propto \frac{1}{T}$$

$$\lambda_m = 2.87 \times 10^{-3} \frac{\text{mK}}{T}$$

4. Define black body radiation?



A perfect black body one which observed and emit all the wavelength of incident radiation on it. The radiation emitted by the black body is known as Black body radiation.

5. Define Matter waves?

Any moving particle is associated with wave nature those waves are called matter waves.

Matter waves are not mechanical waves and electromagnetic wave these are new kind of waves proposed by De-Broglie.

Wavelength $\lambda = h/mv$

6. Explain the Heisenberg uncertainty principle?

According to Heisenberg it is difficult to find the position and momentum simultaneously.

x is the exact position of a particle 'p' is the exact momentum of a particle.

Δx is the error to find the position of a particle, ΔP is the error to find the momentum of a particle. Then the relation between $\Delta x \wedge \Delta P$ from the Heisenberg can be written as

$$\Delta x \cdot \Delta P \geq \frac{h}{4\pi} \vee \Delta x \cdot \Delta P \geq \frac{h}{2}$$
$$\therefore h = \frac{h}{2\pi}$$

7. What is particle and write down its characteristics?

A particle is a small, localized object in physics that can be treated as a single point in space with mass and other physical properties.

Characteristics:

Has mass and may carry charge.

Occupies negligible space (considered a point in ideal cases).

8. What is the statement of Energy quantization by max Planck?

Max Planck proposed that energy is not emitted or absorbed continuously, but in discrete packets called "quanta." According to his theory, the energy E of each quantum is directly proportional to the frequency ν of the radiation, given by the equation $E = h\nu$. This concept explained blackbody radiation and marked the beginning of quantum theory, introducing the idea that atomic and subatomic processes involve quantized energy levels.

9. Write down the types of radiation?



Radiation can be classified into several types based on its nature and how it interacts with matter. The main types of radiation are:

1. Ionizing Radiation

These have enough energy to ionize atoms (remove tightly bound electrons).

Alpha (α) radiation, Beta (β) radiation, Gamma (γ) radiation, X-rays, Neutron radiation

2. Non-ionizing Radiation

These do not carry enough energy to ionize atoms but can excite them or cause heating, Ultraviolet (UV) radiation, Visible light, Microwaves, Infrared (IR) radiation, Radio waves.

10. What is the need for quantum theory?

Quantum theory is needed to explain phenomena that classical physics cannot, such as blackbody radiation, the photoelectric effect, and atomic structure. It accurately describes the behavior of particles at atomic and subatomic scales, where energy is quantized and particles exhibit wave-particle duality, leading to advances in modern physics and technology.

11. Explain planck's radiation law?

- Energy emitted or observed is due to vibration is due to vibration of micro particle (e).
- According to Max-Planck the energy radiated in the form of Quanta. The energy of Quanta $E = h\nu$ in discrete manner.
- The emitted energy is integer multiple of energy of Quanta ($E = h\nu$).

Let us consider black body with large no. of particles N ($N_0, N_1, N_2, N_3, \dots$) with discrete energies E ($0E, 1E, 2E, 3E, \dots$)

N ($N_0, N_1, N_2, N_3, \dots$)

E ($0E, 1E, 2E, 3E, \dots$)

Total no. o Black body

$$N = N_0 + N_1 + N_2 + N_3 + \dots$$

①

Boltzmann relation

$$N_r = N_0 e^{-r E / KT}$$

$r = 0, 1, 2, 3, \dots$

By substituting "r" value in



$$r = 0$$

$$N_0 = N_0 e^{-0.E/KT}$$

$$N_0(1)$$

$$N_0 = N_0$$

$$r = 1$$

$$N_1 = N_0 e^{-1E/KT}$$

$$N_0 e^{-E/KT}$$

$$r = 2$$

$$N_2 = N_0 e^{-2E/KT}$$

N_0, N_1, N_2, N_3 in equation ①

$$N = N_0 + N_0 e^{-E/KT} + N_0 e^{-2E/KT} + N_0 e^{-3E/KT} + \dots$$

$$N_0$$

$$\text{let } x = e^{-E/KT}$$

$$N_0$$

$$\therefore (1 - x)^{-1} = 1 + x + x^2 + x^3 + \dots$$

$$N = N_0(1 - x)^{-1}$$

$$N = \frac{N_0}{(1-x)} \quad \text{②}$$

Energy of the system

$$E = N_0(0E) + N_1(1E) + N_2(2E) + N_3(3E) + \dots$$

$$N_1(1E) + N_2(2E) + N_3(3E) \quad \text{③}$$

Substitute N_1, N_2, N_3 values $\in$ above

$$N_0 e^{-E/KT} (E) + N_0 e^{-2E/KT} (2E) + N_0 e^{-3E/KT} (3E) + \dots$$

$$N_0 e^{-E/KT} (F)$$

$$\text{Let } x = e^{-E/KT}$$

$$N_0 x E$$

$$N_0 x E (1 - x)^{-2}$$



$$E = \frac{N_0 x E}{(1-x)^2} \quad \text{④}$$

$$\text{Average energy of system} = \frac{\text{Total energy of system}}{\text{Total no. of particles}}$$

$$\bar{E} = \frac{\frac{N_0 x E}{(1-x)^2}}{\frac{N_0}{(1-x)}}$$

$$\bar{E} = \frac{N_0 x E}{(1-x)^2} \times \frac{1-x}{N_0}$$

$$\bar{E} = \frac{x E}{(1-x)}$$

$$\bar{E} = \frac{E}{x^{-1} - 1}$$

$$\therefore x = e^{-E/KT}$$

$$\bar{E} = \frac{E}{(e^{hv/KT} - 1)} [\because E = hu]$$

$$\bar{E} = \frac{hv}{e^{hv/KT} - 1} \quad \text{⑤}$$

Black body absorbed and emit all the wavelength (from minimum wavelength to maximum) in the range λ to $\lambda + d\lambda$.

The no. of photons per unit volume

$$\frac{8\pi}{\lambda^4} d\lambda \quad \text{⑥}$$

By multiplying equation 5 with 6 we can get total energy emitted or absorbed by black body.

$$Ed\lambda = \frac{hv}{(e^{hv/KT} - 1) \times \frac{8\pi}{\lambda^4} d\lambda}$$

Equation 7 is known as Planck's law for Black body radiation. It explained blackbody radiation for every wavelength.



12. What is black body radiation? Write a note on laws of Black body radiation?

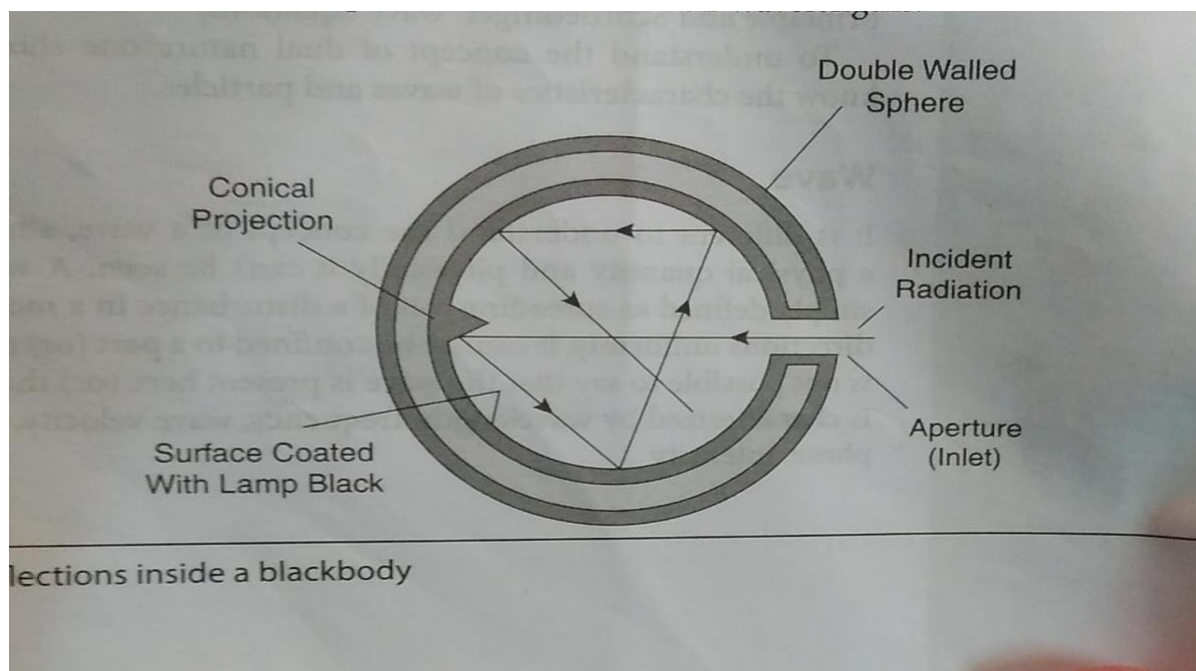
Black body:

A perfect black body one which observed and emitted all the wavelength of incident radiation on it.

Black body radiation:

The radiation emitted by the black body is known as Black body radiation.

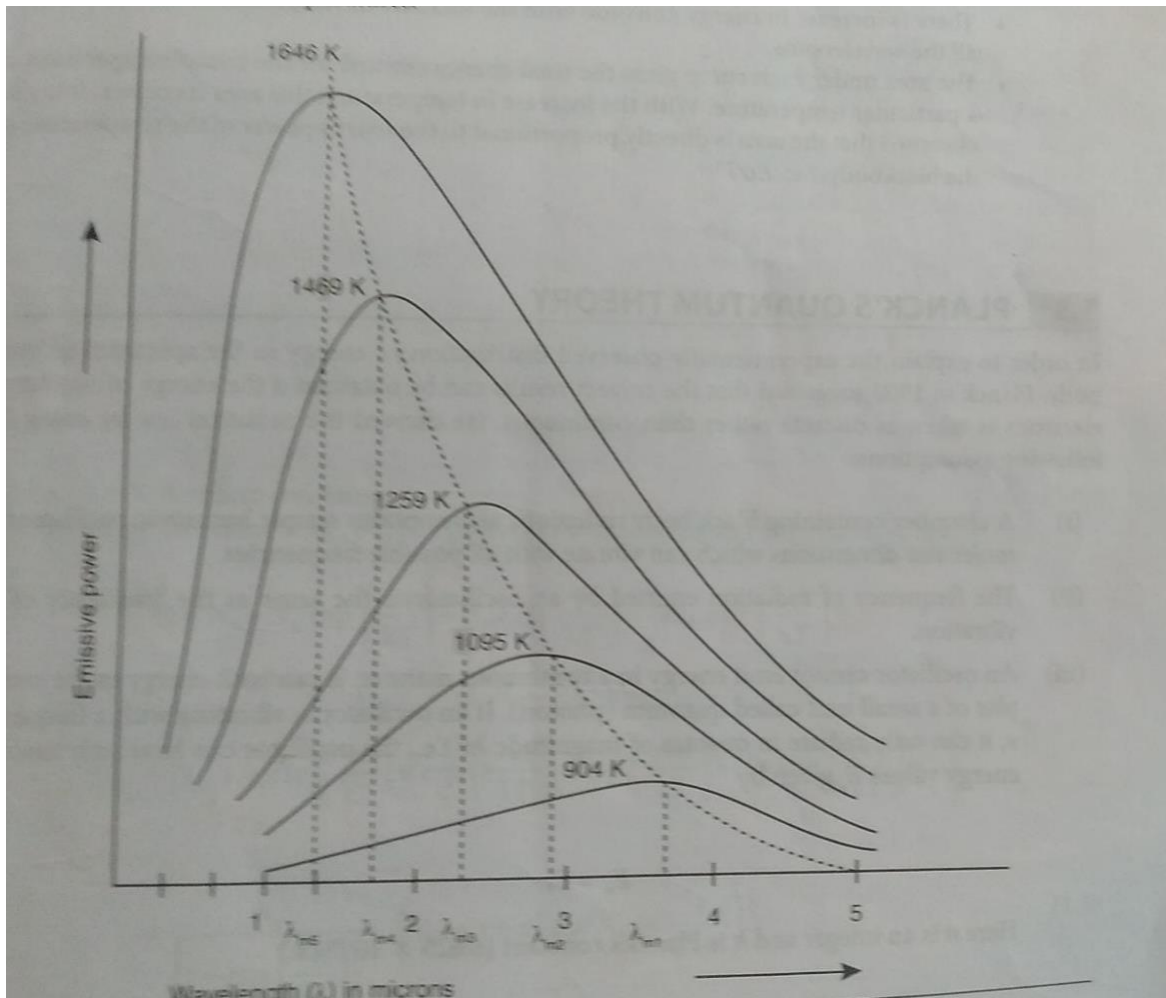
Construction of Black body: Black body is a copper hallow sphere its inner surface is coated with the lamp black; a hole is made on it for entering the radiation. When the radiation entered through this hole into copper hallow sphere due to multiple reflections all the wavelength of radiation is observed. When the copper hallow sphere heated to a particular temperature (or) at thermal equilibrium it will emit all the wavelength of radiation.





Energy spectrum for Black body radiation:

By plotting the graph between wavelength and energy distribution of black body it will be





Energy distribution is not uniform at any given temperature. At a given temperature, with increasing wavelength intensity distribution of black body increases at a particular wavelength reaches maximum (λ_m) behind that intensity decreases.

With increase in temperature (λ_m) (maximum radiation emitted by the black body) decreases. The area of the curve at a temperature will give the radiation emitted by the black body.

Stephen - Boltzmann law:

Stephen-Boltzmann law is the relation between the radiation emitted by the black body to the temperature.

The total radiation emitted by the black body is directly proportional to the fourth power of absolute temperature of black body.

i.e., $E \propto T^4$

$E = \sigma T^4$

$\sigma = 5.67 \times 10^{-8} \text{ w/m}^2 - \text{K}^4.$

Wein's law (Wein's displacement law):

- It explains the shifting of peak wavelength with the temperature.
- The peak wavelength emitted by the black body temperature is inversely proportional to the absolute temperature of black body.

$$\lambda_m \propto \frac{1}{T}.$$
$$\lambda_m = 2.87 \times 10^{-3} \frac{\text{mK}}{T}.$$

Rayleigh-Jean's law

According to Rayleigh-jean's the radiation emitted by the black body at a temperature is directly proportional to absolute temperature of black body and inversely proportional to fourth power of the wavelength.

$$E \propto T$$
$$E(u) \propto \frac{1}{\lambda^4}$$
$$E(u) = \frac{8\pi kT}{\lambda^4}$$



13. Explain de-broglie Hypothesis

De-Broglie proposed the dual nature for the matter. According to De-Broglie matter exhibit both nature particle nature and wave nature.

Matter wave:

Any moving particle is associated with wave nature those waves are called matter waves.

Matter waves are not mechanical wave and electromagnetic wave these are new kind of waves proposed by De-Broglie.

$$\text{Wave length of matter wave } \lambda = \frac{h}{mv} = \frac{h}{p}$$

h = Planck's constant.

$h = 6.825 \times 10^{-34}$ J. Sec, 6.62×10^{-27} erg-sec.

m = Mass of the particle.

v = Velocity of the particle.

De-Broglie wavelength for the matter wave:

De-Broglie calculated the wavelength of matter wave by combining Max-Planck's radiation law and Einstein Mass-Energy relation.

According to Max-Planck's radiation law $E = h\nu$ ①

According to Einstein mass-energy relation $E = mc^2$. ②

From equation 1 and 2

$$h\nu = mc^2 \quad [\because \nu = \frac{c}{\lambda}]$$

$$hc =$$

$$\frac{mc^2}{\lambda} \quad c = 3 \times 10^8 \text{ m/s}$$

$$\lambda = \frac{h}{mc}$$

Particle velocity 'v'

$$\text{The } \lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{p}$$

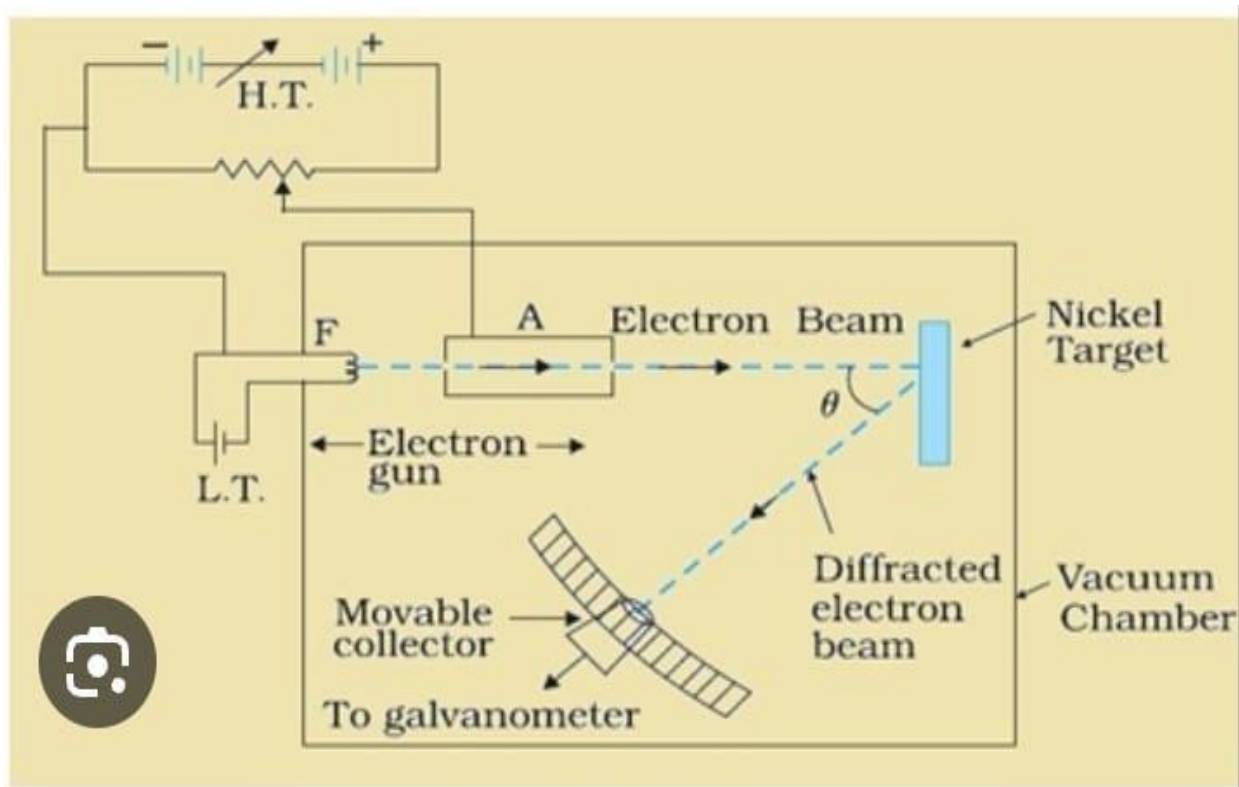
14. Describe the experiment to verify the existence of matter waves?

Davisson and Germer experiment is the practical evidence for the wave nature of the matter.



This was explained by CH. Davisson and LH. Germer

The experimental set up of Davisson and Germer experiment is



1) Electron gun:

Electron gun produces Electron beam it consists of filaments, low tension battery, High tension battery, pin hole cylindrical arrangement to get fine single beam of electron.

When filament gets heated electron gun produces electron beam these electrons are accelerated by providing sufficient potential from High tension batteries.

2) Target set-up (Ni-crystal):

The fast movement of electron beam from electron gun is made incident on Ni-crystal, due to atomic planes of Ni-crystal electrons get diffracted in all possible directions.

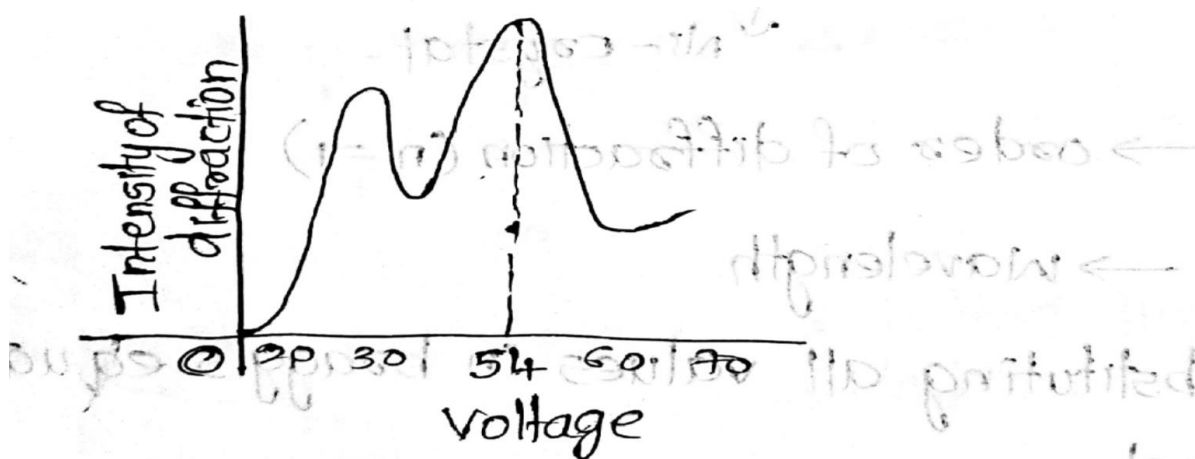
3) Circular scale arrangement:



It is used to collect the diffracted electron beam. It consists of one electron collector which is movable on circular scale, the electron collector is connected to sensitive galvanometer to measure the intensity of diffracted electron beam.

Calculation of wavelength associate from the electron:

When the voltage from High tension battery $V = 54$, The angle between incident beam to diffracted beam is 50° then we will get maximum diffraction pattern (the order of diffraction $n=1$).



The wavelength associate from the electron according to de-broglie

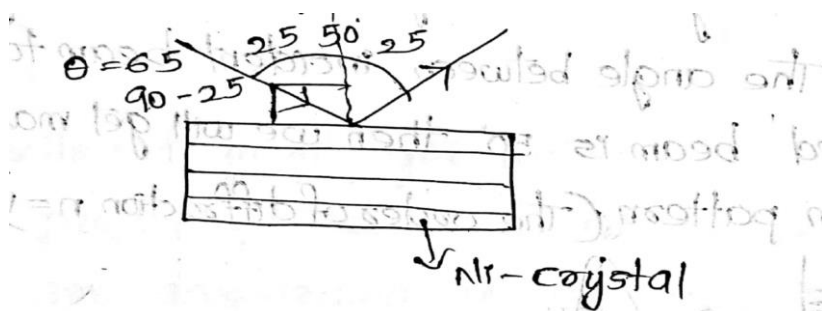
$$\lambda = \frac{12.26}{\sqrt{V}} \text{Å}$$

$$\lambda = \frac{12.26}{\sqrt{54}}$$

$$\lambda = 1.65 \text{Å}$$

The wavelength is associated from Bragg's law for the electron.

Bragg's law $2d \sin \theta = n\lambda$



By substituting all values in Bragg's equation, we get



$$2 \times 0.91 \times 10^{-10} \sin 65^\circ = 1 \times \lambda$$

$$\lambda = 1.66 \text{ \AA}$$

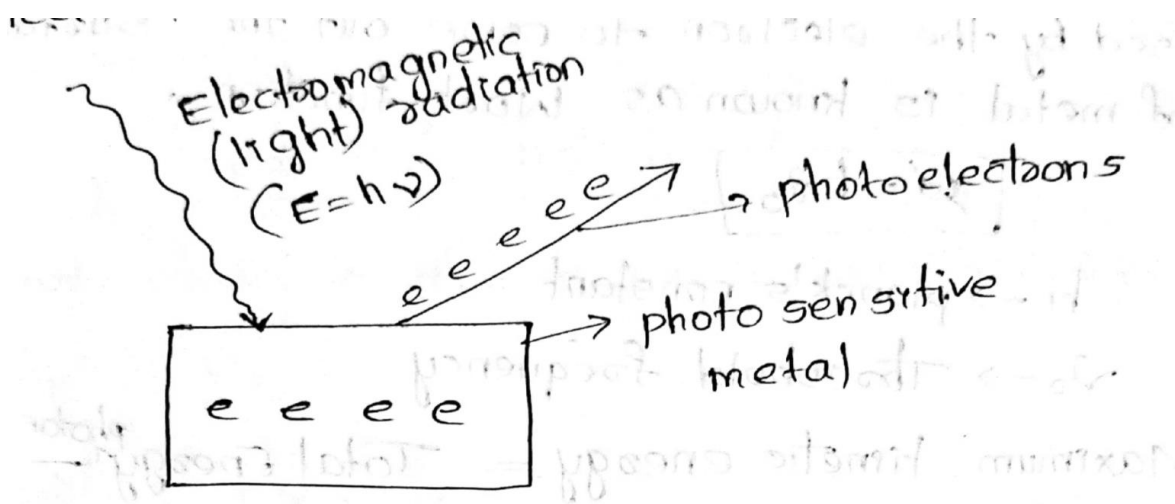
A good agreement between De-Broglie and Bragg's law for the electron

Hence, the wave nature of electron was proved.

15. Define photoelectric effect? With mathematical expression?

When the electromagnetic radiation (light) is incident on a photo sensitive metal if the electrons are emitted/ejected from the surface of metal then that phenomenon is called photo electric effect.

Einstein explained photo electric effect phenomena by using Max-Planck quantum theory.



According to Einstein, energy radiated in the form of photon.

Energy of photon $E = h\nu$

Explanation:

When the electromagnetic radiation is incident on a photosensitive metal photon ($E = h\nu$) collides with the electron, during this collision photon transfer all the amount of energy to electron.

In these total energy part of energy used by the electron comes out the surface of the metal remaining part of energy used by the electron as maximum Kinetic energy with this electron leaves from the metal surface.

Work function:

The minimum amount of energy used by the electron to come out the surface



of metal is known as work function. $\phi = h\nu_0$

$h \rightarrow$ Plank's constant.

$\nu_0 \rightarrow$ Threshold frequency.

Maximum Kinetic energy = Total energy photon - Work function.

$$KE = h\nu - \phi$$

$$\frac{1}{2}mv^2 = h\nu - h\nu_0$$

$$\frac{1}{2}mv^2 = h(\nu - \nu_0)$$

Condition for photo electric effect:

We know mathematical expressions for photo electric effect.

$$\frac{1}{2}mv^2 = h\nu - h\nu_0$$

Kinetic energy is either positive (or) zero.

i.e., $K.E \geq 0$

$$\rightarrow h\nu - h\nu_0 \geq 0$$

$$h\nu \geq h\nu_0$$

$$\nu \geq \nu_0$$

$\nu \rightarrow$ Incident radiation frequency.

$\nu_0 \rightarrow$ Threshold frequency.

Conclusion:

To observe the photo electric effect phenomena incident radiation frequency must be greater than (or) equal to threshold frequency.

16. Derive the expression for the De-Broglie. Derive the time- dependent Schrodinger's wave equation of electron?

De-Broglie calculated the wavelength of matter wave by combining Max-Planck's radiation law and Einstein Mass-Energy relation.

According to Max-Planck's radiation law $E = h\nu$ $\xrightarrow{\text{①}}$

According to Einstein mass-energy relation $E = mc^2$. $\xrightarrow{\text{②}}$

From equation 1 and 2

$$h\nu = mc^2 \quad \left[\because u = \frac{c}{\lambda} \right]$$

$$\frac{hc}{\lambda} = mc^2$$



$$\lambda = \frac{h}{mc} \quad c = 3 \times 10^8 \text{ m/s}$$

Particle velocity 'v'

$$\text{The } \lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{p}$$

De-Broglie wavelength for the electron:

Let us consider an electron with charge 'e' and mass 'm' moving with velocity 'v' applied potential 'V'

The energy gained by the electron due to potential 'V'

$$E = ev \longrightarrow \text{①}$$

The energy gained by the electron due to motion

$$E = \frac{1}{2} mv^2 \longrightarrow \text{②}$$

From 1 and 2

$$ev = \frac{1}{2} mv^2$$

$$mv^2 = 2ev$$

Multiplying with 'm' on both sides

$$m \times mv^2 = m2ev$$

$$m^2 \times v^2 = 2mev \quad mv = p$$

$$p^2 = 2mev$$

$$p = \sqrt{2mev}$$

$$\text{we know that } \lambda = \frac{h}{p}$$

$$\lambda = \frac{h}{\sqrt{2mev}}$$

h - Planck's constant $6.625 \times 10^{-34} \text{ J. sec.}$

m - Mass of e^- $9.1 \times 10^{-31} \text{ Kg.}$

e - Charge of an e^- $1.602 \times 10^{-19} \text{ coulomb.}$

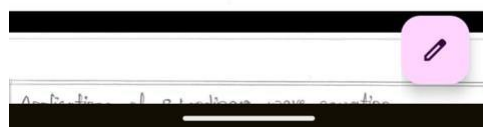
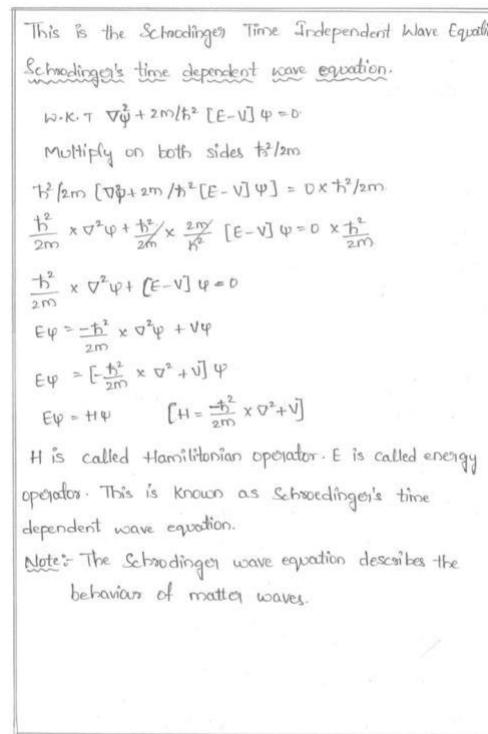
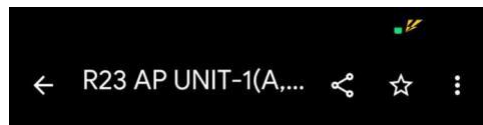
V - Potential difference.

$$\lambda = \frac{6.625 \times 10^{-34} \text{ J. sec}}{\sqrt{2 \times 9.1 \times 10^{-31} \text{ Kg} \times 1.602 \times 10^{-19} \text{ V}}}$$

$$\lambda = \frac{12.26}{\sqrt{V}} \text{ \AA} \quad \text{\AA} = 10^{-8} \text{ cm}, 1 = 10^{-10} \text{ m}$$

Unit for wavelength is $\text{\AA}$.

$$\text{De-Broglie wavelength associate for the energy of the particle } \lambda = \frac{h}{\sqrt{2mE}}$$



17. Derive the time- independent Schrodinger's wave equation of electron ?

Schrodinger wave equation is the fundamental wave equation in Quantum physics to explain the motion of microscopic particles.

To derive the Schrodinger wave equation

Let us consider an electron with mass 'm' moving with velocity 'v' in positive 'x' direction under the potential 'V'.

The general wave equation for the electron can be represented as

$$\psi(x) = A \sin\left(\frac{2\pi}{\lambda}x\right) \quad \text{①}$$

Differentiate above with r to "x".

$$\frac{d\psi(x)}{dx^2} = A \cos\left(\frac{2\pi}{\lambda}x\right) \times \frac{2\pi}{\lambda}$$



Again differentiate above with respect to “x”

$$\frac{d^2\psi(x)}{dx^2} = -A\sin\left(\frac{2\pi}{\lambda}x\right) \cdot \left(\frac{2\pi}{\lambda}\right)\left(\frac{2\pi}{\lambda}\right)$$

$$\frac{d^2\psi(x)}{dx^2} = \frac{-4\pi^2}{\lambda^2} A\sin\left(\frac{2\pi}{\lambda}x\right)$$

From equation 1

$$\frac{d^2\psi(x)}{dx^2} = \frac{-4\pi^2}{\lambda^2} \psi(x)$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2}{\lambda^2} \psi(x) = 0 \text{ (2)}$$

From the De-Broglie

$$\therefore \lambda = \frac{h}{mv}$$

λ substitute in equation 2

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2}{\left(\frac{h}{mv}\right)^2} \psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2 m^2 v^2}{h^2} \psi(x) = 0 \text{ (3)}$$

Total energy of particle = K.E + P.E.

$$E = \frac{1}{2}mv^2 + V$$

$$E - V = \frac{1}{2}mv^2$$

$$2(E - V) = mv^2.$$

Multiply with “m” on both sides.

$$m \times 2(E - V) = m \times mv^2 \quad 2m(E - V) = m^2v^2 \text{ substitute } \in \text{ eqn 3}$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2(2m(E - V))}{h^2} \psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m(E - V)}{h^2} \psi(x) = 0 \text{ (4)}$$

$\psi(x) \rightarrow$ Wave function.

$x \rightarrow$ Propagation of wave.



$m \rightarrow$ Mass of the particle.

$E \rightarrow$ Total Energy of the particle.

$V \rightarrow$ Applied potential.

$h \rightarrow$ Planck's constant.

Equation 4 is known as Schrodinger wave equation (or) Schrodinger 1-D wave equation (or) Time independent wave equation.

$$\text{Since } \therefore k = \frac{h}{2\pi}$$

$$\rightarrow h = k2\pi$$

Substitute in equation 4

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m(E - V)}{k^24\pi^2} \psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m(E - V)}{k^2} \psi(x) = 0$$

This is also Schrodinger wave equation in terms of 'h'

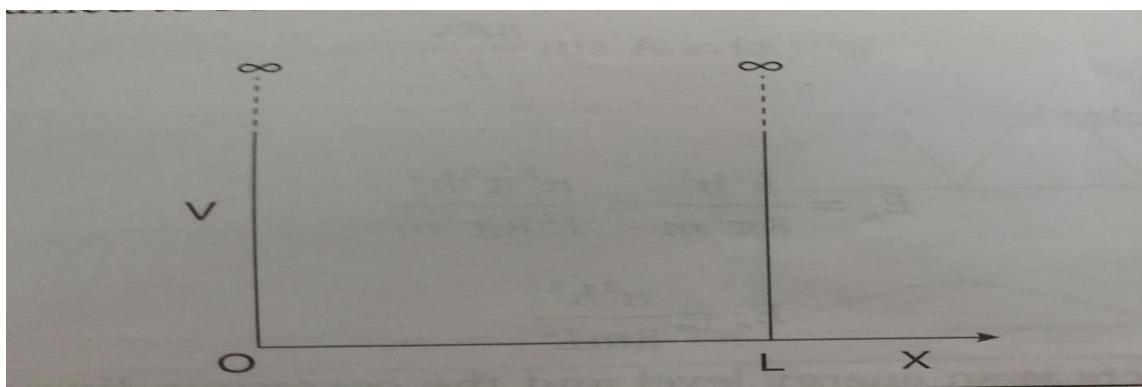
Schrodinger wave equation in 3-Dimension can be represented as

$$\frac{d^2\psi(x)}{dx^2} + \frac{d^2\psi(y)}{dy^2} + \frac{d^2\psi(z)}{dz^2} + \frac{8\pi^2m(E - V)}{h^2} \psi(x, y, z) = 0$$

18. Derive a wave function equation for electron by Trapping in one dimensional potential box?

Let us consider a one-dimensional potential box of width 'a'

Let consider an electron mass is 'm' moving with velocity v in positive x direction.





The motion of microscopic particle is explained by Schrodinger wave equation.

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m(E - V(x))}{h^2}\psi(x) = 0 \quad \text{①}$$

Inside of the box

$$V(x) = 0; 0 < x < a$$

Outside of box

$$V(x) = \infty; x < 0()x > a$$

Apply Schrodinger wave equation inside box of

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m(E - V(x))}{h^2}\psi(x) = 0 [\because V(x) = 0]$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m(E - 0)}{h^2}\psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \left(\frac{8\pi^2mE}{h^2}\right)\psi(x) = 0 [\because K^2 = \frac{8\pi^2mE}{h^2}]$$

$$\frac{d^2\psi(x)}{dx^2} + K^2 \psi(x) = 0 \quad \text{②}$$

The solution of equation 2 from the second order differentiation can be written as

$$\psi(x) = A\sin kx + B\cos kx \quad \text{③}$$

Boundary conditions for 1-D potential box.

At I-Boundary the wave function is 0

$$\text{At } x = 0 \Rightarrow \psi(x) = 0$$

At II-Boundary the wave function 0

$$\text{At } x = a \Rightarrow \psi(x) = 0$$

Apply Boundary condition I & II for equation 3

$$\text{At } x = 0 \Rightarrow \psi(x) = 0 \Rightarrow \psi(0) = A\sin kx + B\cos kx$$

$$0 = A\sin k(0) + B\cos(0)$$

$$\Rightarrow 0 + B = 0$$

$$B = 0$$

$$\text{At } x = a \Rightarrow \psi(x) = 0$$



$$0 = A \sin k(a) + B \cos k(a)$$

$$A \sin ka + B \cos ka (\because B = 0)$$

$$\Rightarrow A \sin ka = 0$$

$$\sin ka = 0$$

$$ka = \sin^{-1}(0)$$

$$\sin^{-1}(\sin \pi) ka = n\pi.$$

$$k = \frac{n\pi}{a} [k \rightarrow \text{wavevector}]$$

$$\therefore k^2 = \frac{8\pi^2 m E}{h^2}$$

Substitute "k" value.

$$\left(\frac{n\pi}{a}\right)^2 = \frac{8\pi^2 m E}{h^2}$$

$$\frac{n^2 \pi^2}{a^2} = \frac{8\pi^2 m E}{h^2}$$

$$E_n = \frac{n^2 h^2}{8ma^2}$$

If $n=1$

$$E_1 = \frac{(1)^2 h^2}{8ma^2} = \frac{h^2}{8ma^2} (\text{Ground state of electron})$$

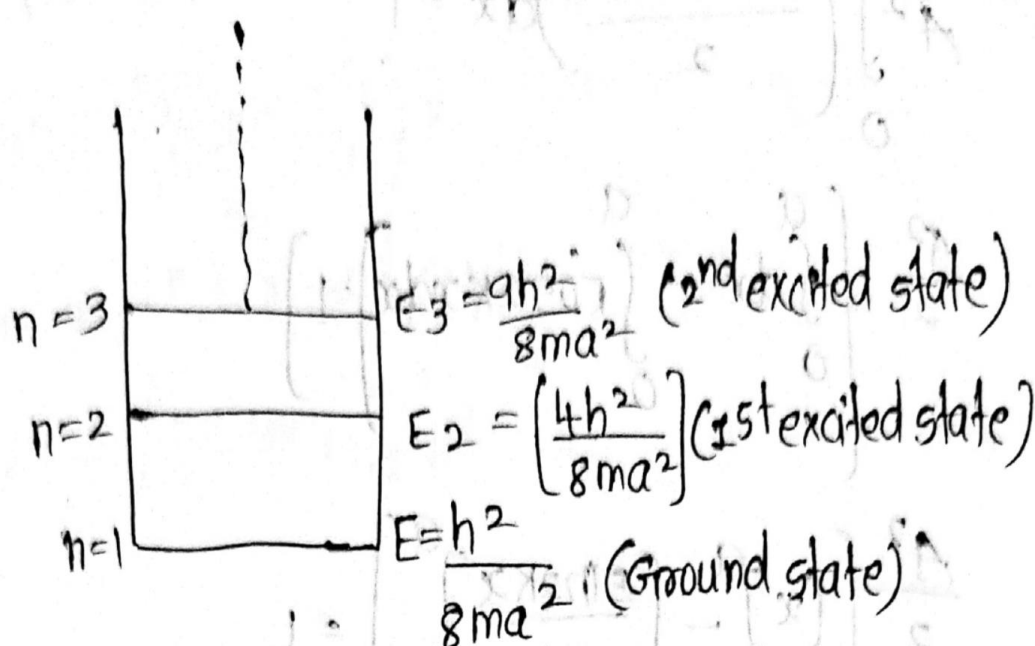
If $n=2$

$$E_2 = \frac{4h^2}{8ma^2} (\text{1st excited state})$$

If $n=3$

$$E_3 = \frac{9h^2}{8ma^2} (\text{2nd excited state})$$

Energy of electron is Quantized and emit in discrete manner.



19. Write a note on Born's interpretation of wave function?

Wave function is a mathematical expression represented in complex number. It has no direct physical meaning.

square of the wave function $[\psi^2(x) = \psi(x)\psi(x)]$

is called probability density

It gives finding the position of a particle in a given region.

If the particle is presented in a given region, then probability density maximum this condition is known as normalization Condition.

Mathematically we can represent as.

$$\int \psi^2(x) dx = 1 [1 - D]$$

$$\int \psi^2(x, y, z) dv = 1 [3 - D]$$



If the particle does not exist in a given region, then probability density minimum this condition is known as orthogonal condition.

Mathematically represented as.

$$\int \psi^2(x)dx = 0[1 - D]$$

$$\int \psi^2(x, y, z)dv = 0[3 - D]$$

UNIT II

1. Write any two assumptions given by Sommerfield?

1. Valence electrons move freely in constant potential within the boundary of metal.
Hence, electrons are trapped in potential boxes.
2. The distribution of electrons in various energy levels occurs as per Pauli's exclusion principle.

Distribution of energy of electrons obeys fermi-dirac distribution.

2. Write any two Merits of classical free electron theory?

1. It explained electrical conductivity, thermal conductivity of metal.
2. It explained Wiedemann - Franz law.
3. It explained optical properties of metal

3. Explain about the Insulators with examples?

The valence band & conduction bands are separated by a very large energy gap (Eg greater or equals to 3eV) At room temp conduction band is empty & valence band is full of electrons. Even at high temperature valence electrons unable to jump into conduction

Eg: Paper, Wood

4. Define Fermi energy level?

The Fermi level is the highest occupied energy level of electrons in a material



at absolute zero temperature. It represents the energy level at which the probability of finding an electron is 50%.

5. Define Semiconductors with two examples?

The valence band and conduction bands are separated with a small energy band gap ($E_g = 1\text{eV}$). At low temperature (0K) free electrons are not available in conduction band hence they like an insulator. The electrical conductivity increases with increasing temperature thus the semiconductors behave as electrical conductors at high temperature. Example: Silicon $E_g = 1.1\text{eV}$
Germanium $E_g = 0.7\text{eV}$

6. What are the assumptions, merits and Demerits of Quantum free electron theory

Quantum free electron theory:

Assumptions:

- It overcomes drawbacks of classical free electron theory.
- Sommerfeld explained them by using Fermi Dirac distribution instead of Maxwell-Boltzmann distribution.
 - 1) Valence electrons move freely in constant potential within the boundary of metal. Hence, electrons are trapped in potential box.
 - 2) The distribution of electrons in various energy levels occurs as per Pauli's exclusion principle.
 - 3) Distribution of energy of electrons obeys fermi-dirac distribution.
 - 4) The energy of electrons is quantized & discrete.

$$i. e., E_n = \frac{n^2 h^2}{8ma^2}$$



Merits:

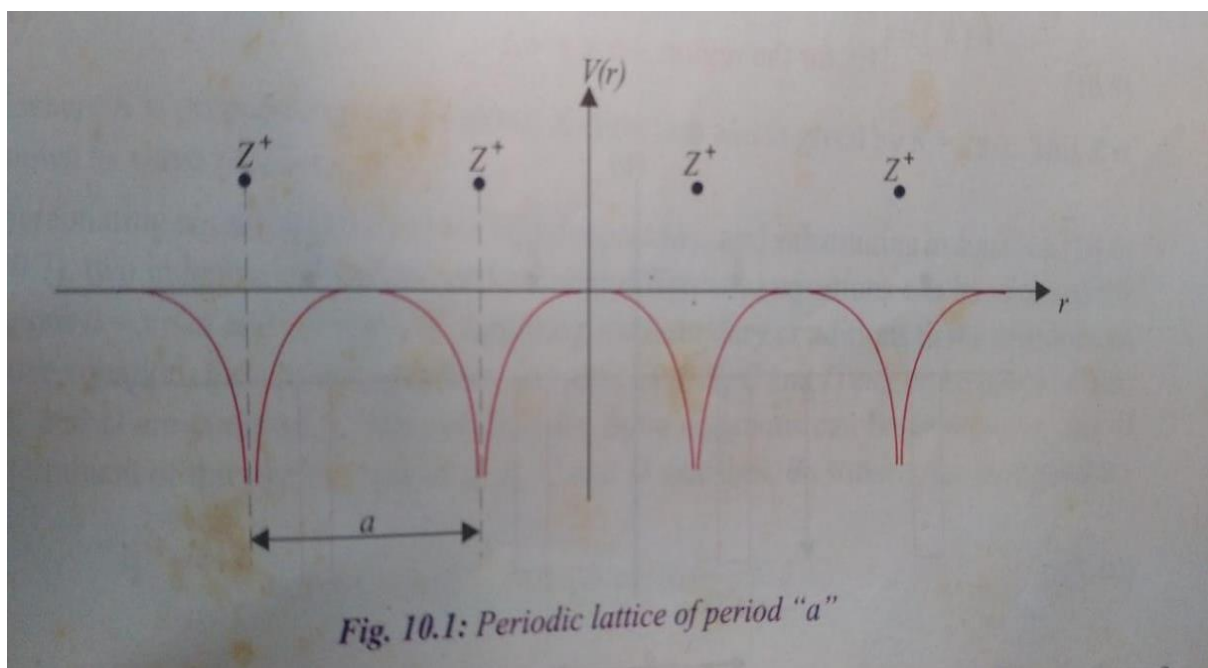
- It explains electrical conductivity and thermal conductivity of metal.
- It explains the variation of electrical conductivity & thermal with temperature.
- It explains photoelectric effects, Compton effect & black body radiation.
- It explains specific heat of electron & susceptibility of para magnetic & ferromagnetic material with the temperature.

De-merits:

- It fails to distinguish between conductors, insulators, semiconductors.
- It fails to explain Hall effect phenomenon.
- According to quantum free electron theory only two electrons exist in fermi energy level, which are responsible for conductivity, but this is not true.

7. Using the Schrodinger time independent wave equation derive the expression for the Bloch equation?

- Let us consider a +ve ion core ('N' no of ions) with periodicity 'a' (distance between any two ions).
- Let's consider an electron moving freely between the positive ion core, when the electron at the site of +ve ion the potential is minimum ($V=0$), when electrons between the +ve ions the potential is maximum (V_0).





- The motion of electrons is explained by schrodinger equation.

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m[E - V(x)]}{h^2} \psi(x) = 0 \quad \text{--- (i)}$$

The potential for single ion $V(x) = V(x + a)$ --- (ii)

Substitute (ii) in (i)

$$\frac{d^2\psi(x)}{dx^2} + 3\pi^2m \frac{[E - V(x + a)]}{h^2} + \psi(x) = 0 \quad \text{---(iii)}$$

- From the block the solution of equation (iii) can be written as multiplication of plane wave & periodicity.

$$\psi(x) = u_k(x)e^{ikx} \quad \text{--- (iv)}$$

- If we consider 'N' no. of atom, then potential

$$V(x) = V(x + Na)$$

$$u_k(x) = u_k(x + Na) \text{-----} v$$

Then the solution from bloch.

$$\psi(x + Na) = u_k(x + Na)e^{ik(x + Na)}$$

From equation number "v"

$$\psi(x + Na) = [u_k(x)]e^{iKx} \cdot e^{iKNa}$$

$$= u_kx \cdot e^{iKx} \cdot e^{iKNa} \Rightarrow \text{from equation(iv)}$$

$$\psi(x + Na) = \psi(x) \cdot e^{iKNa} \quad \text{--- (vi)}$$

This is known as Bloch equation (or) theorem complex

Conjugate of equation (vi)

$$\psi^* \quad (x + Na) = \psi^* \cdot e^{-iKNa} \quad \text{--- (vii)}$$

Multiply (vi) & (vii)

$$\psi(x + Na) \cdot \psi^*(x) = \psi(x)\psi^*(x)e^{-iKNa}$$

$$|\psi(x + Na)|^2 = |\psi(x)|^2$$

i.e., probability density for electrons is equal to entire crystal.



8. Give the brief introduction of the Drude and Lorentz theory?

This was proposed by Drude and Lorentz. According to classical free electron theory metal consists of free electrons which are responsible for electrical conductivity. These electrons follow classical laws.

Assumptions: In Metal large no. of free electrons move freely in all possible directions.

These electrons behave as gas molecules in containers which obeys kinetic theory of gases. In the absence of field, the energy associated with electron $E = \frac{3}{2} KT$ which is equal to kinetic energy $\frac{3}{2} kT = \frac{1}{2} mv^2$.

Metal consists of positive ion core in fixed position and electron move freely collides with positive ion core (or) free electron (or) with the metal boundary.

These collisions are perfectly elastic collision. Hence, conductivity is due to free electrons.

The electron velocity obeys Maxwell-Distribution velocity. The potential energy of electrons is always of constant value.

If electric field is applied the electron gets accelerated, the velocity of electron opposite direction of electric field this velocity is called drift velocity (V_d).

Merits: It explained Ohm's law. It explained electrical conductivity, and the thermal conductivity of metal. It explained Wiedemann - Franz law. It explained optical properties of metal.

De-Merits: It fails to explain electrical conductivity & thermal conductivity of semiconductors & Insulators.

It fails to explain the variation of electrical conductivity, thermal conductivity^[2] at low temperature. It fails to explain photoelectric effect, black body radiation and specific heat of electron.

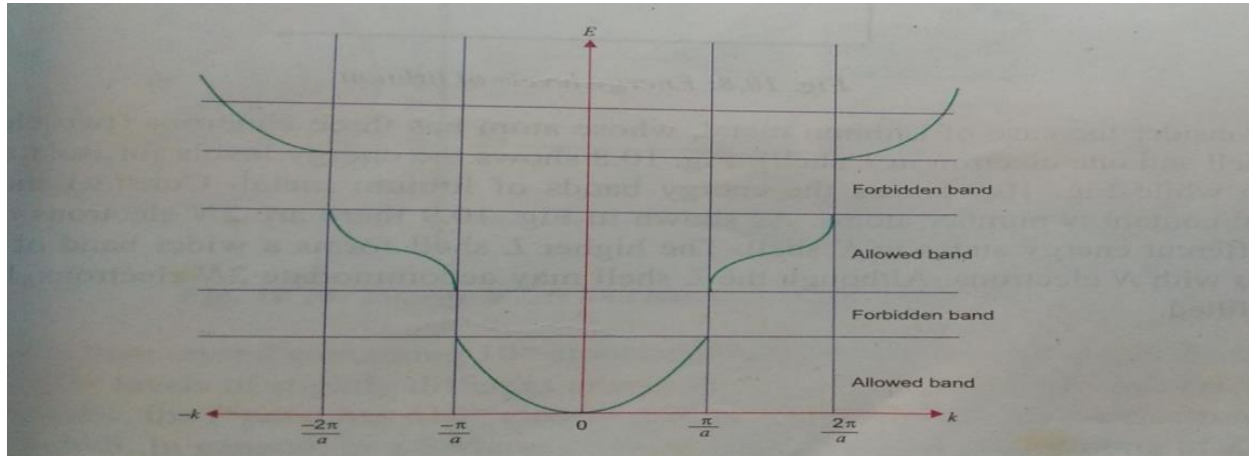
It fails to explain the susceptibility of paramagnetic material and Ferromagnetic material based on temperature.

9. Draw the E-K Diagram and Define the Brillouin Zone and forbidden zone?

E – k diagram by using $P \sin \alpha \cos \alpha + \cos \alpha = \cos k a$ we can plot the curve (graph) energy as a function of wave vector



In above equation right hand side term ($\cos ka = \pm 1$) for the values of $k = \pm h\pi / a$ Hence, discontinuity occurs in energy spectrum consist of allowed region and forbidden region.



If $n = 1$, $k = \pm h\pi / a$ then allowed region extended $-\pi / a$ to π / a called first Brillouin zone.

If $n = 2$, $k = \pm 2\pi / a$ i. e., $k = -2\pi / a, -\pi / a, 0$ and $\pi / a, 2\pi / a$ called second Brillouin zone.

In an **E-k diagram**, the Brillouin Zone defines the range of wave vectors (k) within which the electron energy bands are plotted.

The x-axis represents the wave vector (k), related to the momentum of electrons.

The first Brillouin Zone typically spans from $-\pi/a$ to $+\pi/a$ (in 1D), where a is the lattice constant.

The edges of the Brillouin Zone (e.g., $\pm\pi/a$) are high-symmetry points where energy band gaps may occur due to Bragg reflection of electron waves.

In the diagram:

Brillouin Zones appear as regions between these symmetry points. Energy bands are typically shown folding back at the Brillouin zone boundaries.



Forbidden Energy Band Gap in E–k Diagram:

In an E–k diagram, the forbidden energy band gap is a vertical energy range between two bands where no electronic states exist.

It appears between the top of the valence band and the bottom of the conduction band.

The y-axis (energy E) shows this gap clearly as a region where there are no allowed energy levels

for electrons.

The gap often occurs at the Brillouin zone boundary (e.g., at $k = \pm\pi/a$) due to the periodic potential of the crystal.

10. What are the assumptions and merits of classical free electron theory?

Assumptions: In Metal large no. of free electrons move freely in all possible directions.

These electrons behave as gas molecules in containers which obeys kinetic theory of gases. In the absence of field, the energy associate with electron $E = \frac{3}{2} KT$ which is equal to kinetic energy $\frac{3}{2}kT = \frac{1}{2}mv^2$.

Metal consists of positive ion core in fixed position and electron move freely collides with positive ion cove (or) free e electron (or) with the metal boundary.

These collisions are perfectly elastic collision. Hence, conductivity is due to free electrons.

The electron velocity obeys Maxwell-Distribution velocity. The potential energy of electron is always constant value



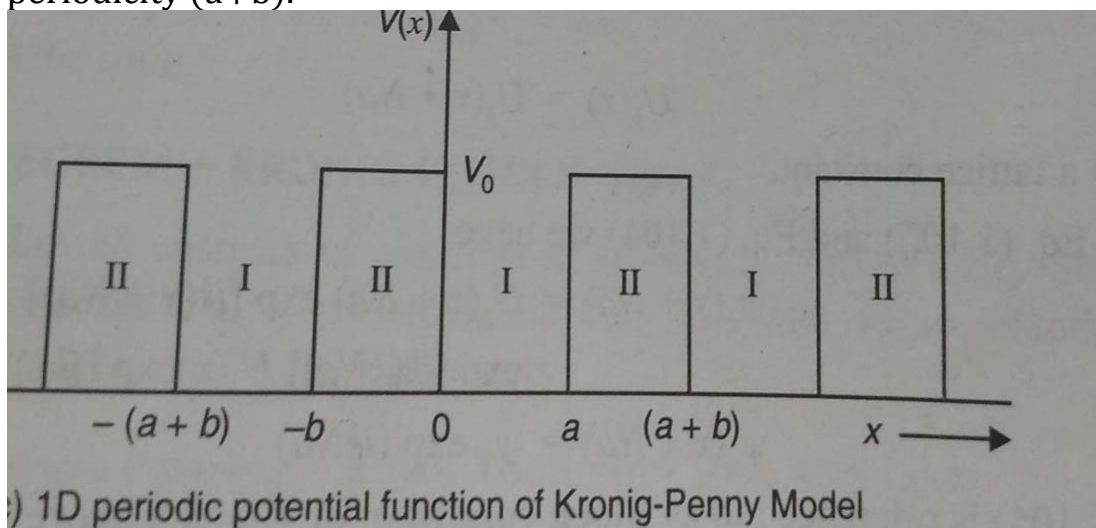
If electric field is applied the electron gets accelerated, the velocity of electron opposite direction of electric field this velocity is called drift velocity (V_d).

Merits: It explained Ohm's law. It explained electrical conductivity and thermal conductivity of metal. It explained Wiedemann - Franz law. It explained optical properties of metal.

11. Explain the Kronig-Penny model for the electron in periodic potential?

They proposed periodic potential.

They approximated periodicity to a 1D rectangular potential with the periodicity $(a+b)$.



The motion of electron is given by Schrodinger wave equation.



$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m[E - V(x)]}{h^2} \psi(x) = 0 \quad \text{--- (i)}$$

Potential:

In region-I $V(x) = 0 \quad 0 < x < a$

region-II $V(x) = V_0 \quad -b < x < 0$

By applying Schrodinger equation in region-I.

$$\begin{aligned} \frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m[E - 0]}{h^2} \psi(x) &= 0 \\ \frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2mE}{h^2} \psi(x) &= 0 \\ \frac{d^2\psi(x)}{dx^2} + \alpha^2 \psi(x) &= 0 \quad \text{--- (ii)} \end{aligned}$$

Apply Schrodinger wave equation region-II.

$$\begin{aligned} V(x) &= V_0, \quad -b < x < 0 \\ \frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m[E - V_0]}{h^2} \psi(x) &= 0 \\ \frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m[V_0 - E]}{h^2} \psi(x) &= 0 \\ \frac{d^2\psi(x)}{dx^2} - \beta^2 \psi(x) &= 0 \quad \text{--- (iii)} \\ \beta^2 &= \frac{8\pi^2m(V_0 - E)}{h^2} \end{aligned}$$

The solution of equation (ii) & equation (iii) from Bloch can be written as

$$\psi(x) = u_k(x)e^{iKx} \quad \text{--- (iv)}$$

after applying boundary conditions by mathematical expression written as

$$\frac{\beta^2 - \alpha^2}{2\alpha\beta} \sin\alpha a \sinh(\beta b) + \cos\alpha a \cosh(\beta b) = \cos k(a + b) \quad \text{--- (v)}$$

While increasing potential of the barrier then width of potential barrier decreases.

$$V_0 \Rightarrow \infty \Rightarrow b = 0$$

$$\text{Then } \sinh(\beta b) \rightarrow \beta b$$

$$\cosh(\beta b) \rightarrow 1$$

Substitute in equation (v)

$$\frac{\beta^2 - \alpha^2}{2\alpha\beta} \sin\alpha a (\beta b) + \cos\alpha a (1) = \cos k(x) \quad \text{--- (vi)}$$

We know that

$$\begin{aligned} \beta &= \frac{8\pi^2m[V_0 - E]}{h^2}; \alpha^2 = \frac{3\pi^2mE}{h^2} \\ \beta^2 - \alpha^2 &= \frac{8\pi^2m(V_0 - E)}{h^2} - \frac{8\pi^2mE}{h^2} \\ &= \frac{8\pi^2m}{h^2} (V_0 - E - E) \\ &= \frac{8\pi^2m}{h^2} (V_0 - 2E) \end{aligned}$$

$$V_0 \gg E$$



$$\beta^2 - \alpha^2 = \frac{8\pi^2 m V_0}{h^2} \quad \text{substitute in (vi)}$$

$$\frac{8\pi^2 m V_0}{h^2 \cdot \alpha \beta} \beta b \sin \alpha a + \cos \alpha a = \cos ka$$

$$\frac{h^2 \cdot \alpha \beta}{4\pi^2 m V_0} b \sin \alpha a + \cos \alpha a = \cos ka$$

$$\frac{h^2 \cdot \alpha}{4\pi^2 m V_0 b a} \sin \alpha a + \cos \alpha a = \cos ka$$

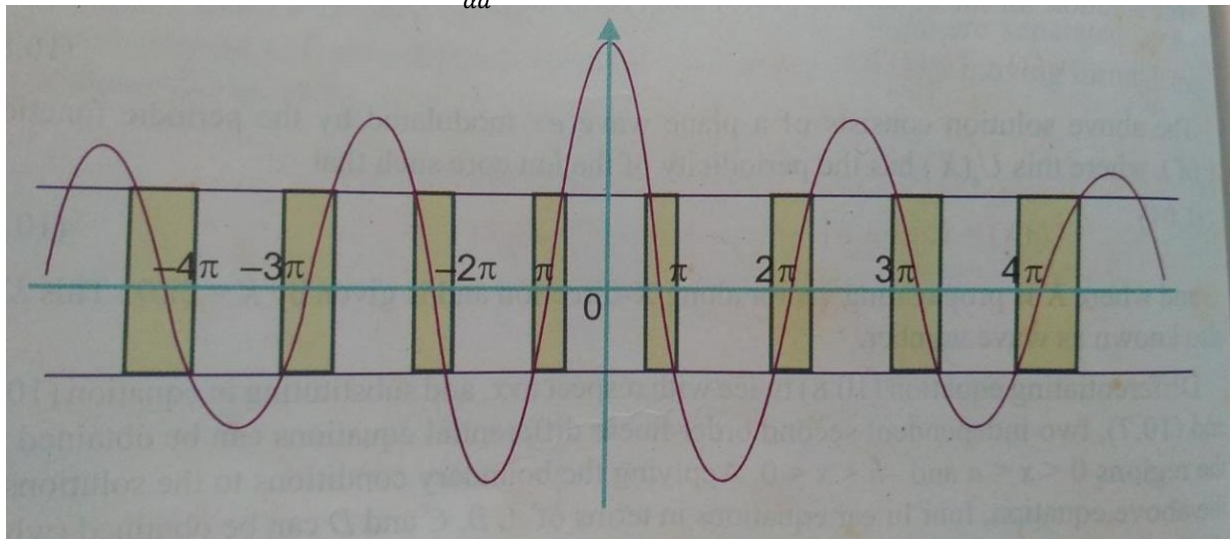
$$\left[\frac{h^2 \alpha a}{4\pi^2 m V_0 b} \right] \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

$P \rightarrow$ Scattering power (or) Barrier strength.

$$\frac{P \sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka \quad \text{--- (vii)}$$

➤ It will give attractions between electron and the +ve electron core.

By plotting graph between $\frac{P \sin \alpha a}{\alpha a} + \cos \alpha a$ and αa then graph will be



➤ From the graph electron characterized in an energy level corresponding to particular value of αa . This leads the existence of allowed region and forbidden region decreases.

➤ If P is very large ($P \rightarrow \infty$) P is barrier strength.

from equation (vii)

$$\frac{\sin \alpha a}{\alpha a} = 0$$

$$\sin \alpha a = 0$$

$$\alpha a = \sin^{-1}(0)$$

$$\alpha a = \sin^{-1}(\sin n\pi)$$

$$\alpha a = n\pi$$

$$\alpha = \frac{n\pi}{a}$$

$$\frac{P \sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

$$[\because P \rightarrow \infty]$$

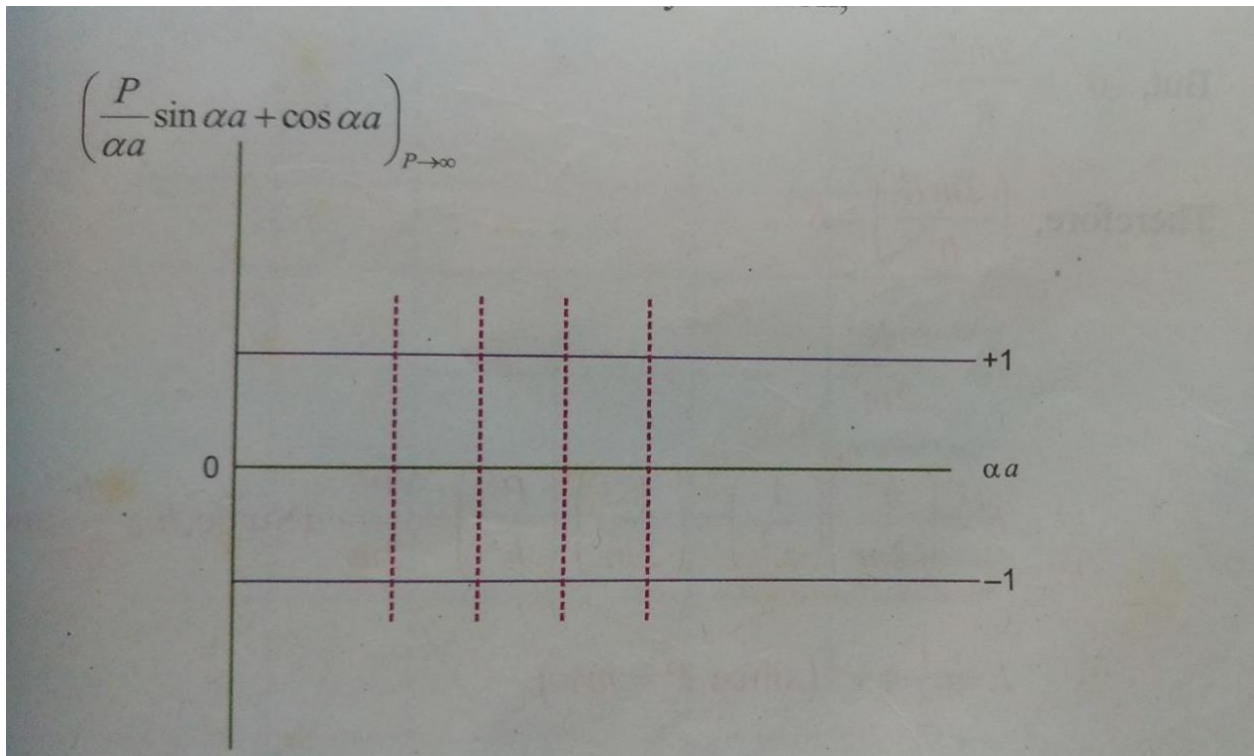


$$\Rightarrow \alpha^2 = \frac{n^2 \pi^2}{8\pi^2 m E} = \frac{a^2}{h^2} = \frac{n^2 h^2}{8ma^2}$$

$$E = \frac{n^2 h^2}{8ma^2}$$

i.e., energy is quantized for when barrier strength is more.

➤ with increase in barrier strength (P) width of allowed region decreases.



➤ P is very small $P \rightarrow 0$

$$\frac{P \sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

$$\cos \alpha a = \cos ka$$

$$\alpha a = ka \quad [\because P = 0]$$

$$\alpha = k$$

$$\alpha^2 = k^2$$

$$\frac{8\pi^2 m E}{h^2} = \frac{4\pi^2}{\lambda^2}$$

$$\frac{2mE}{h^2} = \frac{1}{\lambda^2}$$

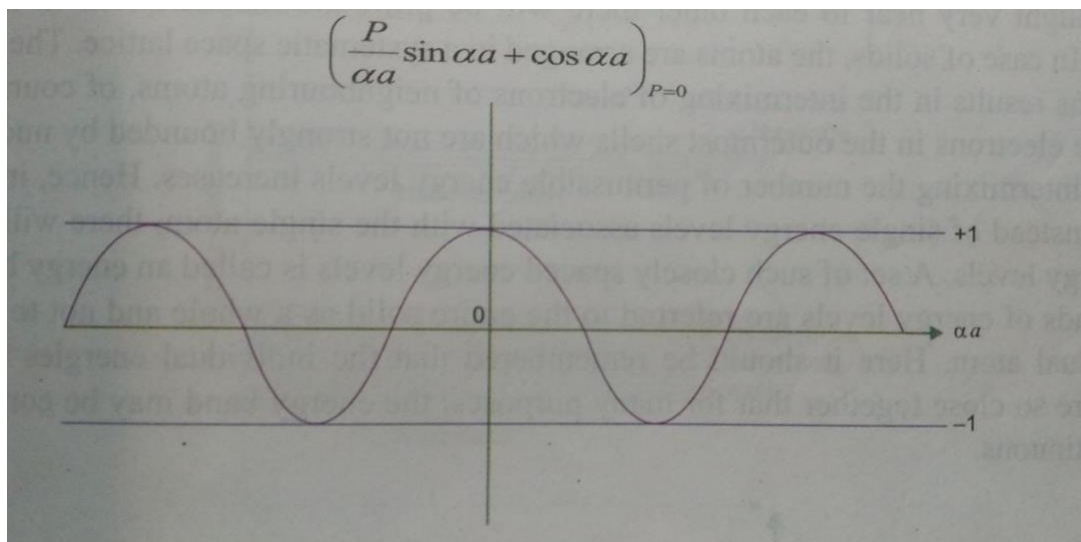
$$[\because k = \frac{2\pi}{\lambda}]$$

$$[\because \lambda = \frac{h}{mv}]$$

$$\frac{2mE}{h^2} = \frac{m^2 u^2}{h^2}$$

$$E = \frac{1}{2} m v^2$$

i.e., the total energy of the electron is from the kinetic energy, potential energy is zero then the particle is free particle then the energy spectrum.



i.e., electrons are allowed in all energy band this support classical free electron theory.

12. Write the Demerits of classical free electron theory?

De-Merits: It fails to explain electrical conductivity & thermal conductivity of semiconductors & Insulators.

It fails to explain the variation of electrical conductivity, thermal conductivity at low temperature. It fails to explain photoelectric effect, black body radiation and specific heat of electron.

It fails to explain the susceptibility of paramagnetic material and Ferromagnetic material based on temperature.

13. Give a brief explanation about conductors, semiconductors and insulators?

Let us consider two identical atoms when these atoms are brought together, first their outermost orbits overlap & interact then the energy levels corresponding to the outermost orbit split into two.

When the three atoms are brought together the outermost orbit split into three energy levels. In the same way 'N' no. of atoms is brought together the outermost orbit split into 'N' energy levels "

The collection of 'N' outermost orbit energy levels forms an energy band is known as "valence band". If the atoms are very closer the next innermost orbits also overlap, they also form an energy bar. But here the innermost bands have no physical significance. The bands corresponding to the outermost orbit valence orbit is called "valence band" and next permitted band is known as conduction band. At 0K valence electrons occupy the valence band and conduction band is empty.



Classification of solids: Based on the energy gap, solids are classified into insulators, semiconductors and conductors.

Conductors: The valence band and conduction bands overlapped with each other, so energy gap (E_g) is zero. At room temperature free electrons already exist in conduction band. Hence solids (metals) are electrical as well as thermal conductors. The electrical resistivity increases with increasing temperature in conductors

Semiconductors: The valence band and conduction bands are separated with a small energy band gap ($E_g = 1\text{eV}$). At low temperature (0K) free electrons are not available in conduction band hence they like an insulator. The electrical conductivity increases with increasing temperature thus the semiconductors behave as electrical conductors at high temperatures.

Example: Silicon $E_g = 1.1\text{eV}$ Germanium $E_g = 0.7\text{eV}$

Insulators: The valence band & conduction bands are separated by a very large energy gap (E_g greater or equals to 3eV) At room temp conduction band is empty & valence band is full of electrons. Even at high temperature valence electrons are unable to jump into conduction band.

UNIT III

1. Give the Introduction of Quantum Computing?

A normal (classical) computer stores and processes information using bits, which can be either 0 or 1. Everything your phone, laptop, or any computer does comes from billions of bits flipping between 0 and 1. A quantum computer works differently—it uses quantum bits, or qubits, which can be 0, 1 or both at the same time.

2. Give brief explanation about Superposition?

Superposition is a unique property of quantum systems where a quantum bit (qubit) can exist in multiple states simultaneously.

Unlike a classical bit, which must be either 0 or 1, a qubit can be in a state that is both 0 and 1 at the same time until it's measured.

Think of it like a coin spinning in the air—until you look (measure), it's both heads and tails at once.

Superposition is what allows quantum computers to process many possibilities all at the same time, vastly increasing their potential computational power for certain tasks.

When you measure (read) a qubit, the superposition collapses, and you find it in one of its possible states.



This concept is often illustrated by the famous "Schrödinger's cat" thought experiment and the double-slit experiment.

In the latter, a single particle, like a photon, goes through two slits simultaneously and creates an interference pattern—showing that it's a superposition of going through both slits at once.

3. Write the notation of Bra-Ket?

Bra notation is written as $\langle\psi|$ (pronounced "bra psi").

A bra is the complex conjugate transpose of a ket. Mathematically, it acts as a row vector that can be paired with a ket to compute an inner product.

In mathematical terms, if $|\psi\rangle$ is a ket, its corresponding bra is $\langle\psi|$.

The bra is a function that maps a ket (vector) to a complex number and is used for measuring overlaps between quantum states.

4. Give brief explanation about Entanglement?

Entanglement is another fundamental concept in quantum mechanics.

It's a special relationship between two or more qubits, where the state of one qubit instantly affects the state of the other, no matter how far apart they are.

If two qubits are entangled, measuring one immediately gives you information about the other,

even if they're separated by huge distances. I

n quantum computing, entanglement allows for linked qubits that work together in ways classical

bits can't.

This property is crucial for the power and complexity of quantum algorithms and for secure quantum communication.

5. Write two advantages of Quantum computing?

Faster Processing: Quantum computers can solve certain complex problems much faster than classical computers, such as factoring large numbers or simulating quantum systems.

Parallelism: Quantum bits (qubits) can exist in multiple states simultaneously (superposition), allowing quantum computers to perform many calculations at once.

6. Write the differences of Classical & quantum bits?

Classical Bits: A classical bit is the basic unit of information in traditional computers. It can only be in one of two states: 0 or 1, like a basic light switch being either off or on.



Computers use billions of these bits to store and process all their data and instructions. Classical bits follow the rules of ordinary physics (classical physics) and are manipulated by logic gates like AND, OR, and NOT.

Quantum Bits (Qubits) A quantum bit, or qubit, is the fundamental unit in quantum computers. Qubits can be 0, 1, or both at the same time (through a property called superposition).

Because of this, qubits can process lots of information at once, making quantum computers very powerful for certain tasks.

Qubits also show entanglement, where the state of one qubit can depend on another, even at a distance.

They operate under the rules of quantum mechanics, not classical physics, and are manipulated by quantum logic gates.